Predictive Code-Switching Detection Project

Course: Machine Learning Final Project (60% of grade)
Timeline: 14 weeks
Dataset: SwitchLingua (420K samples, 12+ language pairs)

What is Code-Switching?

Code-switching is when bilingual speakers alternate between two languages in conversation:

- **Example:** "I'm going to the supermercado to get groceries" (English → Spanish)
- **Example:** "أنا تابعت خبر ترامب، honestly الموضوع معقد" (Arabic → English)

This is NOT a language deficiency—it's a complex, rule-governed linguistic phenomenon.

The Problem

Traditional Approach (Detection): Models identify language switches AFTER they happen in complete sentences.

Our Goal (Prediction): Predict at token position t whether token $t+1$ will be a language switch—like a real-time chatbot or keyboard app would need.

Challenge: We only have the "prefix" (context so far), not the full sentence.

Our Two Tasks (Multitask Learning)

Task 1: Switch Prediction (Binary Classification)

- **Question:** Will the next token be a language switch?
- **Output:** Yes/No (0 or 1)

Task 2: Duration Prediction (3-Class Classification)

- **Question:** If a switch happens, how long will it last?
- **Output:**
    - Small (1-2 tokens)

- Medium (3-6 tokens)
- Large (7+ tokens)

Why both? Multi-tasking helps the model learn the "rhythm" of language switching patterns.

Technical Approach

Model Architecture

- **Backbone Models:** Compare XLM-RoBERTa vs. mBERT/Unicoder
- **Structure:** Shared Encoder → Two Classification Heads (one for each task)
- **Key Feature: Causal Mask** to prevent future information leakage

Training Strategy

- **Shifted Labels:** Train at position t using label from position $t+1$
- **No Cheating:** Model can only see past context, not future tokens

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Success Metric: "Universality"

We're not just measuring accuracy—we're measuring **consistency across languages**.

Universality = Low Variance (σ)

- Calculate F1-score for each of 12 language pairs
- Compute standard deviation (σ) across all pairs
- **Best model = Lowest σ** (most consistent performance)

Example:

- Model A: Some languages F1=0.9, others F1=0.5 → High variance ❌
- Model B: All languages F1=0.7-0.75 → Low variance ✓ **More Universal**

Dataset: SwitchLingua

- **Size:** 420K samples

- **Languages:** 12+ pairs (Hindi-English, Spanish-English, Arabic-English, Cantonese-English, etc.)
- **Quality:** Linguistically validated, natural conversations
- **Access:** Free from Hugging Face: `Shelton1013/SwitchLingua_text`

Project Timeline

Weeks	Task
1-4	Data preprocessing & label engineering (create shifted labels)
5-7	Build model architecture (custom heads + causal mask)
8-10	Training & hyperparameter tuning
11-14	Evaluation, universality analysis, final paper

What Each Team Member Should Know

Data Engineering (Weeks 1-4)

- Parse SwitchLingua dataset
- Tokenize code-switched sentences
- Create shifted labels for both tasks
- Split data by language pairs

Model Development (Weeks 5-7)

- Implement causal masking
- Build dual classification heads
- Set up training pipeline
- Compare XLM-RoBERTa vs. mBERT

Evaluation (Weeks 8-14)

- Calculate F1-scores per language pair
- Compute variance (σ) for universality metric

- Analyze which model handles "small" vs. "large" switches better
- Write final paper with findings

Expected Findings

- Different architectures may specialize in different switch types
- XLM-RoBERTa likely better at cross-lingual embeddings
- Duration prediction difficulty: Small lexical switches vs. Large clausal switches
- Some language pairs may be inherently harder to predict

Deliverables

1. ✅ Preprocessed dataset with shifted labels
2. ✅ Trained models (multiple architectures)
3. ✅ Performance analysis across 12 language pairs
4. ✅ Universality audit (variance analysis)
5. ✅ Final research paper

Key Takeaway

We're building a **real-time language switch predictor** that works consistently across multiple language pairs—solving a problem that traditional detection models can't handle.

Real-world applications: Bilingual keyboards, chatbots, translation tools, language learning apps.